PEDAGOGIA SCHOOL & COLLEGE

Computer (PBA) (Morning)

SSC – II

(FBISE 1st Annual Examination 2026)

(Answer Key)

TIME ALLOWED: 2:40 Hours **Total Marks: 40**

Note: The questions of sections B and C are to be answered on the separately provided answer book. Use supplementary answer sheet i.e. Sheet – B if required. Write your answers neatly and legibly.

SECTION- A (Marks 40)

Attempt the following questions briefly.

Q.1	**a. Debug and correct the following code to produce the given output:**
	<script>
	var sum;
	var a=5;
	var b=6;
	var sum, prod;
	sum = a + b;
	prod = a * b;
	document.writeln("Sum=" + sum + "
	");
	document.write("Product=" + prod);
	</script>
		05

	OUTPUT: Sum = 11 Product = 210	**05**
	b. Write a JavaScript program to calculate the total marks of students. 1. Declare three variables to store the marks of three subjects (English, Urdu, Math). 2. Let marks be: English = 60, Urdu = 75, Math = 80. 3. Calculate total marks. 4. Display the total marks.	**05**

	OR
	a. Write a JavaScript code that displays the numbers from 1 to 10 using loop.
	
	b. What does the following JavaScript code print?
	<script>
	document.write("This program displays the values of array
	");
	var a = [3, 5, 3, 2, 1];
	document.write("Array values are: " + a);
	</script>
		05

	Design a webpage displaying a Student Result table using HTML tags. The page should include the following elements: **a. Create the HTML structure and add the title of the page using <title> tag: Student Result**	**02**
	b. Inside the <body> tag: 1. Add main heading: **"Report of Student Result"** using <h1> tag. 2. Bold the above heading using <b> tag. 3. Add a brief paragraph to introduce the student.	**1+1+1**

	4. Add a table listing 3 students record with student name and marks. **Student Name** | **Marks** Amina Khan | 400 Ahmed Ali | 350 Sarah Akhtar | 410	**05**
Q.3	**Write a CSS code to apply the following styles to the table in Question 2:** 1. Set the text color (foreground color) of the table content. 2. Set the alignment of all table contents to the left. **b. Identify the organism shown in diagram 3.2.** **Diagram 3.2**	**2+2**
Q.4	Desing a concept map to represent the video game development. OR Develop a flowchart to find the area of triangle when height and base are given.	**05**
Q.5	a. Label the following devices:  b. Classify the following software as system or application software: 	**2.5+2.5**
Q.6	Consider the following business idea for Online Education Platform and prepare the business plan provided below: An online company name "Edu Hub Pakistan" that offers interactive video courses, live classes, and digital learning resources for students and professionals. The system provides course tracking certificates and easy mobile access. Following are the key features: * Recorded and live learning options. * Low cost subscription plans. * Certificate and progress tracking dashboard. Company Name: 01 Company's Slogan: 01 Executive summary: (in one line) 01 Company Objectives: (Any two) 02 Potential Customers / Market: (Any two) 01	**06**

PEDAGOGIA SCHOOL & COLLEGE

Computer (PBA) (Evening)

SSC – II

(FBISE 1st Annual Examination 2026)

(Answer Key)

TIME ALLOWED: 2 Hours **Total Marks: 40**

Note: The questions of sections B and C are to be answered on the separately provided answer book. Use supplementary answer sheet i.e. Sheet – B if required. Write your answers neatly and legibly.

SECTION- A (Marks 40)

Attempt the following questions briefly.

Q.1

a. Debug and correct the following code to produce the given output: — **05**

```
<script>
VAR sub 1=55, sub2=67, sub3=78;
var total, percentage;
total = subl+sub2 sub;
percentage=(total/300*100
document.writ ("Total="+total+ "<br>") ;
document.write (""Percentage="+percentaget "B");
<script>
```

OUTPUT:
Total=200
Percentage=66.666% — **05**

b. Write a JavaScript program to calculate the total marks of students.

1. Declare two variables and assign them values (like a=5, b=10)
2. Declare a third variable, addition, and calculate the addition of two numbers.
3. Display the result of addition using document.write ().

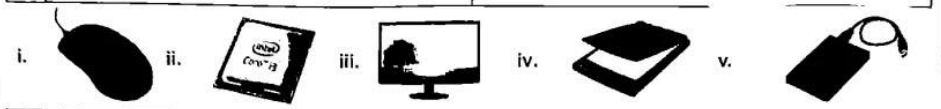
05

OR

a. Write a JavaScript code that displays the numbers from 10 to 20 using for loop.

b. What does the following JavaScript code Prints? — **05**

```
<script>
var marks=[ ];
marks[0]= 73;
marks[1]= 80;
marks[2]= 90;
marks[3]= 88;
</script>
document.write ("Marks of student are:"+marks) ;
document.write ("This program displays the marks of students <br>") ;
```

	Design a webpage displaying a Student Result table using HTML tags. The page should include the following elements: **a. Create the HTML structure and add the title of the page using <title> tag: "Online Book Store"** **Student Result** **b. Inside the <body> tag:** 1. Add Main heading "Introduction to Book Store" using <h2> tag. 2. Underline the above heading using <u> tag. 3. Add image "book.jpg" of bookshelf using <img> tag. 4. add a table listing 3 books with title and author. Book Title Author Computer Mr. Shahid Maths Ms. Ayesha English Mr, Ahmed	**02** **1+1+1** **05**
	School: Admission Open (Pre 9th & 9th)	
Q.3	**Write a CSS code to apply the following styles to the table in Question 2:** 1. Set the text color (foreground color) of the table content to yellow. 2. Set the text alignment of all table content to the left.	**2+2**
Q.4	Desing a concept map to issue a book in a library. Include checks for book availability and the maximum number of books a user can borrow. **OR** Develop a flowchart to find the largest of three unequal numbers.	**05**
Q.5	a. Label the following devices:  b. Classify the following software as system software or application software: 	**2.5+2.5**
Q.6	Consider the following business idea for Online Freelancing Platform and prepare the business plan provided below: A web based company name "Skill Development Pakistan" that connects clients and freelancers for secure payments, transparent rating system, and easy project management tools. Following are the key features: Provides opportunities for skilled professional and students. Supports multiple payment methods (PayPal, Easy-paisa, Jazz Cash etc.) °Offers service tracking and feedback system.	**06**

MORNING GROUP

FBISE SSC-II Computer Science PBA | 1st Annual 2026 | Complete solutions

Fakhar STEM Sphere | Questions, reasoning, outputs and exam guidance. Morning solutions appear first; Evening solutions follow. All OR options are included. These answers follow the supplied school-prepared copies; apparent source errors are explained.

Q1(a) Debug the sum and product program

Question: Correct the given program to display Sum = 11 and Product = 210.

Source discrepancy: The supplied code has a = 5 and b = 6. Their sum is 11, but their product is 30. The repeated declaration of var sum is unnecessary, but is not a JavaScript syntax error. To match the displayed product, the solution below assumes an additional factor c = 7, used only in the product.

```
<script>
var a = 5;
var b = 6;
var c = 7;
var sum, prod;
sum = a + b;
prod = a * b * c;
document.write("Sum = " + sum + "<br>");
document.write("Product = " + prod);
</script>
```

Output and reasoning

```
Sum = 11
Product = 210
```

Sum = 5 + 6 = **11**; product = 5 * 6 * 7 = **210**. The + operator adds numbers and also joins text to a value in the display statement. The * operator multiplies;
 inserts a line break.

Exam tip: State the additional-factor assumption. If the actual question requires only the product of 5 and 6, the answer is 30. If it requires the sum and product of 5, 6 and 7, the answers are 18 and 210.

Q1(b) Calculate total marks

Question: Store English = 60, Urdu = 75 and Math = 80 in three variables. Calculate and display their total using a fourth variable.

```
<script>
var english = 60, urdu = 75, math = 80;
var totalMarks;
totalMarks = english + urdu + math;
document.write("Total Marks = " + totalMarks);
</script>
```

Output: Total Marks = 215. **Reason:** 60 + 75 + 80 = 215.

Exam tip: Use totalMarks or total_marks. Variable names cannot contain spaces.

MORNING GROUP

Q1 alternative questions | Both OR parts solved

Q1 - OR (a) Display numbers from 1 to 10

Question: Write JavaScript code to display numbers from 1 to 10 using a loop.

```
<script>
var i;
for (i = 1; i <= 10; i++) {
    document.write(i + "<br>");
}
</script>
```

Output

```
1
2
3
4
5
6
7
8
9
10
```

Loop component	Purpose
i = 1	Starts the counter at 1.
i <= 10	Includes all values up to and including 10.
i++	Increases the counter by one after each repetition.
document.write(...)	Displays the current number, followed by a line break.

After 10 is displayed, i becomes 11. The condition 11 <= 10 is false, so execution leaves the loop. There are 10 repetitions.

Exam tip: Using i < 10 would omit 10. Keep all three loop components inside the parentheses, separated by semicolons.

Q1 - OR (b) Predict the array output

Given code in this attached copy:

```
<script>
document.write("This program displays the values of array <br>");
var a = [3, 5, 3, 2, 1];
document.write("Array values are: " + a);
</script>
```

Correct output

```
This program displays the values of array
Array values are: 3,5,3,2,1
```

Reason: The first statement prints its text and a line break. Joining an array to a string converts the array into comma-separated values. The square brackets and spaces between array entries are not printed.

Exam tip: Array indices start at zero. Here a[0] = 3 and a[4] = 1. Copy the values from the actual paper: earlier scans in this conversation used a different array.

MORNING GROUP

Q2 | Student Result webpage

Q2 Design a Student Result webpage

Requirements: Title Student Result; main heading Report of Student Result using <h1>; bold heading using <b>; an introductory paragraph; and a table containing the three given student records.

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Result</title>
</head>
<body>
    <h1><b>Report of Student Result</b></h1>
    <p>
        This report presents the results of three students.
        Their names and obtained marks are given below.
    </p>
    <table border="1">
        <tr>
            <th>Student Name</th>
            <th>Marks</th>
        </tr>
        <tr>
            <td>Amina Khan</td>
            <td>400</td>
        </tr>
        <tr>
            <td>Ahmed Ali</td>
            <td>350</td>
        </tr>
        <tr>
            <td>Sarah Akhtar</td>
            <td>410</td>
        </tr>
    </table>
</body>
</html>
```

Expected table

Student Name	Marks
Amina Khan	400
Ahmed Ali	350
Sarah Akhtar	410

Reason: <title> sets the browser-tab title; <body> holds visible content; <h1> creates the heading; <b> makes it bold; <p> creates a paragraph. A <tr> creates a row, <th> creates a heading cell and <td> creates a data cell.

Exam tip: Use four table rows: one heading row and three student rows. Include <b> because the question requests it, even though h1 is normally bold by default. Close paired tags neatly.

MORNING GROUP

Q3 CSS | Q4 concept map

Q3 Style the Student Result table

Question: Set the foreground colour of the table content and align all table contents to the left. The attached copy omits the colour; the earlier original-paper scan specifies green. Green is used below.

```
<style>
table {
    color: green;
}
th, td {
    text-align: left;
}
</style>
```

Placement: Insert the style block inside <head> in the Q2 HTML document.

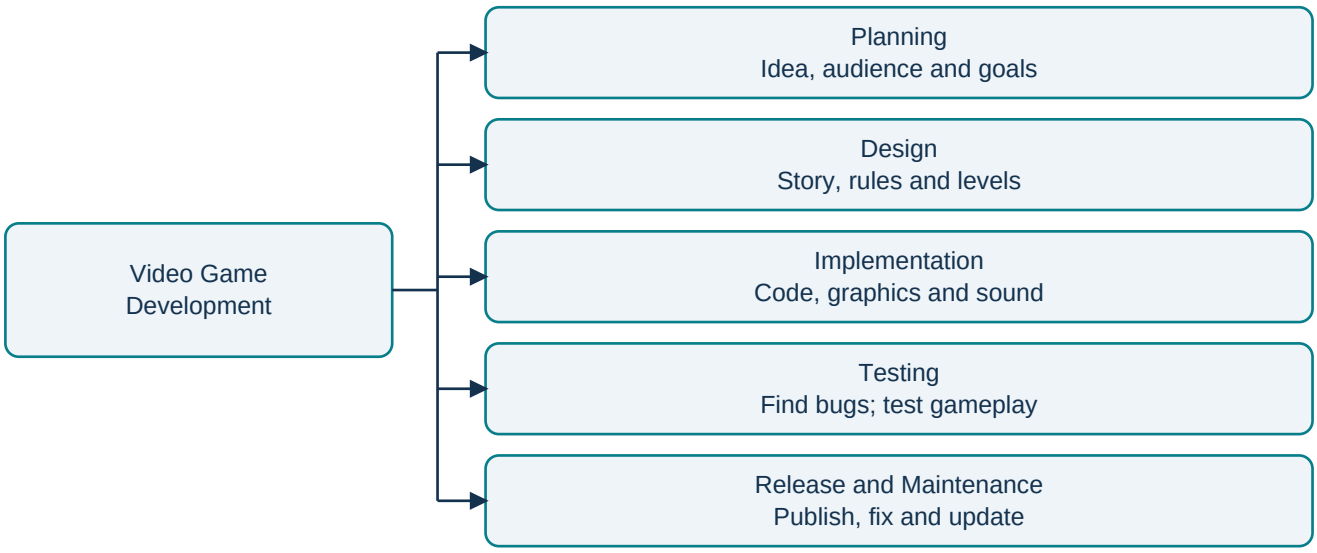
Property / selector	Meaning
color: green;	Makes the foreground text green.
th, td	Selects both heading cells and data cells.
text-align: left;	Aligns the content inside these cells to the left.

Exam tip: Use the actual paper's specified colour. If it asks for background colour, use background-color; if it asks for text colour, use color.

Unrelated printed line: The copy also asks to identify an organism in Diagram 3.2, but no such diagram is supplied. An organism cannot be identified from the provided page.

Q4 Concept map: video game development

Question: Design a concept map representing video game development.



Reason: Planning establishes the idea and audience. Design develops the rules, story and levels. Implementation combines code, graphics and sound. Testing finds problems. Release and maintenance deliver the game and improve it after publication.

Exam tip: Draw a central topic with labelled branches and short details. A paragraph alone does not replace the requested concept map.

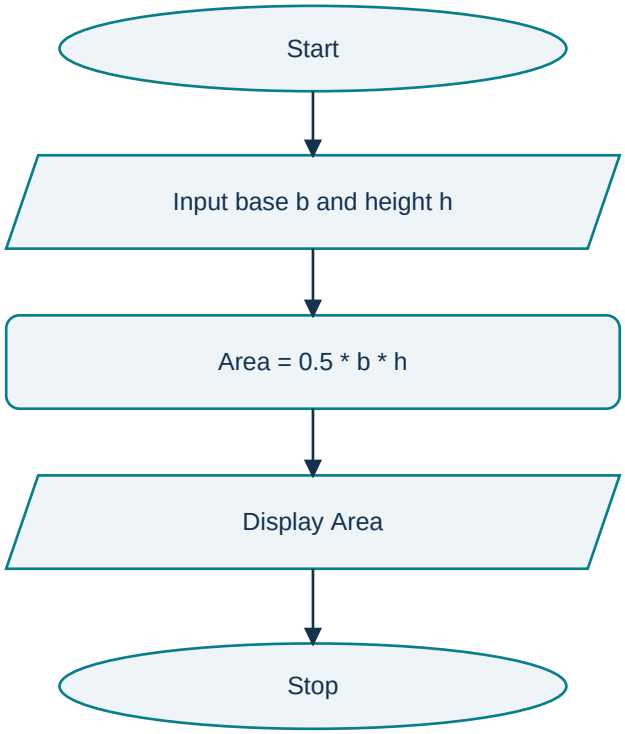
MORNING GROUP

Q4 OR flowchart | Q5(a) device identification

Q4 - OR Find the area of a triangle

Question: Draw a flowchart to find the area when the base and height are given.

Formula: Area = (base * perpendicular height) / 2 = 0.5 * b * h.



Example: If b = 10 cm and h = 6 cm, Area = 0.5 * 10 * 6 = **30 square centimetres**.

Flowchart component	Correct symbol
Start / Stop	Oval
Input / Display	Parallelogram
Calculation	Rectangle

Exam tip: The height must be perpendicular to the base. Connect all symbols with arrows and put the calculation inside the processing rectangle.

Q5(a) Label the devices

Label	Device	Identification / function
i	Motherboard	Main circuit board with slots, connectors and a processor socket.
ii	Mouse	Input device used to point, click and select.
iii	USB flash drive / pen drive	Portable storage with a USB connector.
iv	Optical disc (CD/DVD)	Circular storage medium read using a laser.
v	Printer	Output device producing a hard copy on paper.

Exam tip: The disc image is too unclear to distinguish a CD from a DVD reliably; Optical disc (CD/DVD) is a sound identification. Number answers to match the pictures.

MORNING GROUP

Q5(b) classification | Q6 business plan

Q5(b) System software or application software?

Label / software	Classification	Reason
i. Microsoft Word	Application software	Creates and edits documents.
ii. AVG Antivirus	System software - utility	Detects and removes malware to maintain system security.
iii. Windows	System software - operating system	Manages hardware, files and application execution.
iv. Google Chrome	Application software	Browses websites.
v. Android	System software - operating system	Manages mobile hardware and applications.

Concept: System software operates or maintains the computer. Application software performs tasks for the user. For AVG, write **System software (antivirus utility)**.

Q6 Business plan: Edu Hub Pakistan

Scenario: An online education company provides live and recorded courses, digital resources, low-cost subscriptions, mobile access, certificates and progress tracking.

Required component	Exam-ready answer
Company name	Edu Hub Pakistan
Slogan	Learn Anywhere, Achieve Everywhere.
Executive summary (one line)	Edu Hub Pakistan provides affordable online courses, live classes, recorded lessons and learning resources for students and professionals.
Objective 1	Provide accessible, affordable education through low-cost subscriptions and mobile access.
Objective 2	Improve knowledge and skills through interactive courses, progress tracking and certificates.
Potential customer / market 1	School, college and university students seeking academic support.
Potential customer / market 2	Working professionals seeking further learning and skill development.

Reason: The objectives match the platform's affordability, accessibility, teaching and tracking features. The customer groups match the students and professionals named in the scenario.

Exam tip: Use the exact company name, one short summary sentence, two distinct objectives and two customer groups. A slogan should be short and memorable.

Before moving to the Evening paper

Check that every program has declarations, correct calculations and output. In HTML, check nesting and closing tags. Practise all OR questions, but follow the actual examination instructions when choosing which alternative to attempt.

EVENING GROUP

FBISE SSC-II Computer Science PBA | 1st Annual 2026 | Complete solutions

Q1(a) Debug the total and percentage program

Question: Correct the program using marks 55, 67 and 78. The target total is 200; the printed percentage is 66.666%. Assume each subject is out of 100.

```
<script>
var sub1 = 55;
var sub2 = 67;
var sub3 = 78;
var total, percentage;
total = sub1 + sub2 + sub3;
percentage = (total / 300) * 100;
document.write("Total = " + total + "<br>");
document.write("Percentage = " + percentage + "%");
</script>
```

Calculation and output

Total = 55 + 67 + 78 = **200**.
Percentage = (obtained marks / maximum marks) * 100 = (200 / 300) * 100 = **66.6666...%**.

```
Total = 200
Percentage = 66.66666666666666%
```

Corrections: Use lowercase var; remove spaces from sub1; use sub1 + sub2 + sub3; close the percentage expression; correct document.writ to document.write; repair quotes and concatenation; display "%"; close the block with </script>.

Precision note: The printed 66.666% is truncated. Correct rounding to three decimal places is **66.667%**. To display a rounded result, replace the percentage display statement with:

```
document.write("Percentage = " + percentage.toFixed(3) + "%");
```

If exactly 66.666% is explicitly required, truncate this positive percentage instead:

```
var truncated = Math.floor(percentage * 1000) / 1000;
document.write("Percentage = " + truncated + "%");
```

Exam tip: Do not alter the subject marks to force the printed decimal. Keep the correct percentage formula and distinguish rounding from truncation.

Q1(b) Add two numbers

Question: Despite its marks-related heading, the detailed instructions ask for a = 5, b = 10 and a third variable named addition.

```
<script>
var a = 5, b = 10;
var addition;
addition = a + b;
document.write("Addition = " + addition);
</script>
```

Output: Addition = 15. **Reason:** 5 + 10 = 15. Follow the specific numbered instructions.

EVENING GROUP

Q1 alternative questions | Both OR parts solved

Q1 - OR (a) Display numbers from 10 to 20

```
<script>
var i;
for (i = 10; i <= 20; i++) {
    document.write(i + "<br>");
}
</script>
```

Output: The numbers 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 and 20, each on a separate line.

Reason: Start at 10, continue while i <= 20, and increase by one after each repetition. At i = 21 the condition is false. The loop runs **20 - 10 + 1 = 11 times**.

Exam tip: The endpoints are included. There are 11 numbers, not 10.

Q1 - OR (b) Predict the marks-array output

Printing error: The supplied page places </script> before the two document.write statements. As printed in HTML, those statements are outside the JavaScript block and do not execute as JavaScript. Their source text may appear as page text. The intended array-output answer assumes the closing tag is moved after both statements.

Intended code after repairing the script boundary

```
<script>
var marks = [];
marks[0] = 73;
marks[1] = 80;
marks[2] = 90;
marks[3] = 88;
document.write("Marks of student are:" + marks);
document.write("This program displays the marks of students <br>");
</script>
```

Intended output with the given statement order

```
Marks of student are:73,80,90,88This program displays the marks
of students
```

Display note: The two lines above are wrapped only to fit this page. There is **no programmed line break between 88 and This**. The
 occurs only at the end of the second statement.

Reason: Array indices begin at 0. The four stored values are 73, 80, 90 and 88. String concatenation displays the array as comma-separated values without square brackets.

Optional improvement for a two-line result

```
document.write("Marks of student are: " + marks + "<br>");

Marks of student are: 73,80,90,88
This program displays the marks of students
```

Exam tip: For output prediction, preserve the given statement order and line breaks. State the script-boundary defect before giving the intended answer; do not silently repair it.

EVENING GROUP

Q2 | Online Book Store webpage

Q2 Design the Online Book Store webpage

Requirements: Title Online Book Store; heading Introduction to Book Store using <h2>; underline using <u>; image book.jpg; and a table with three book titles and authors. These specific instructions take precedence over the leftover Student Result wording in the copy.

```
<!DOCTYPE html>
<html>
<head>
    <title>Online Book Store</title>
</head>
<body>
    <h2><u>Introduction to Book Store</u></h2>
    <img src="book.jpg" alt="Bookshelf">
    <table border="1">
        <tr>
            <th>Book Title</th>
            <th>Author</th>
        </tr>
        <tr>
            <td>Computer</td>
            <td>Mr. Shahid</td>
        </tr>
        <tr>
            <td>Maths</td>
            <td>Ms. Ayesha</td>
        </tr>
        <tr>
            <td>English</td>
            <td>Mr. Ahmed</td>
        </tr>
    </table>
</body>
</html>
```

Expected table

Book Title	Author
Computer	Mr. Shahid
Maths	Ms. Ayesha
English	Mr. Ahmed

Reason: <title> sets the browser-tab title; <h2> gives the required heading level; <u> underlines it; src identifies the image file and alt provides alternative text. The table has one heading row and three data rows.

Exam tip: Write book.jpg exactly. When running this relative path, place the image in the same folder as the HTML file. The img element has no closing </img> tag. Use h2, not h1, for this question.

EVENING GROUP

Q3 CSS | Q4 library concept map

Q3 Yellow table text with left alignment

Question: Apply yellow foreground text and align all table content to the left. Insert the following style block inside <head> in Q2.

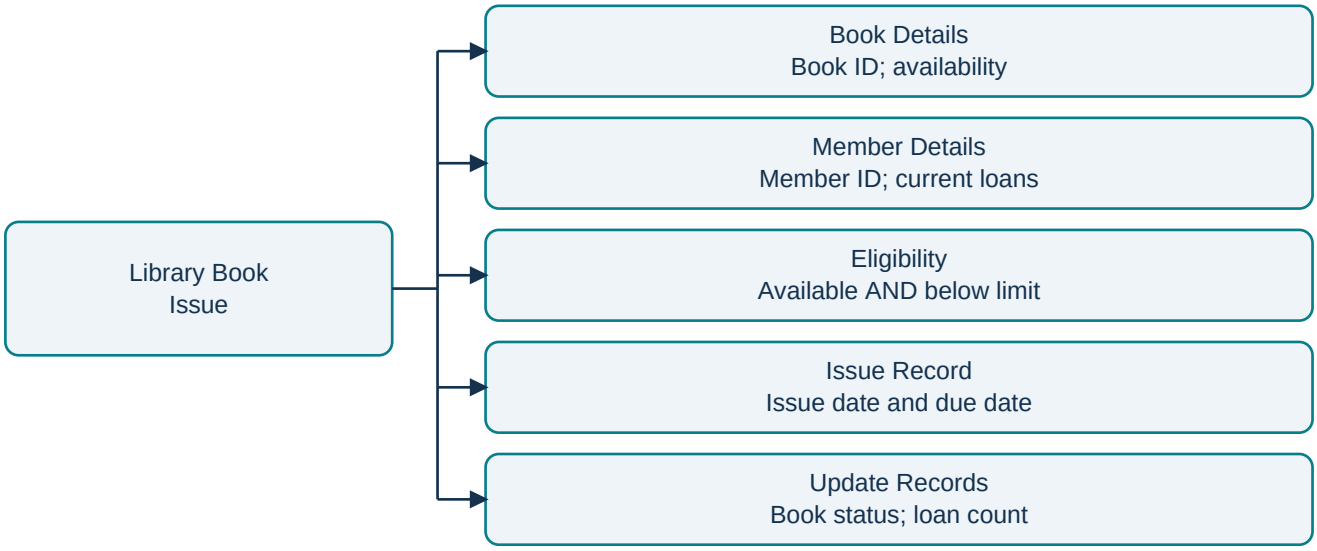
```
<style>
table {
    color: yellow;
}
th, td {
    text-align: left;
}
</style>
```

Reason: color controls foreground text. The selector th, td covers both heading and data cells, including headings that browsers normally centre. text-align: left aligns their contents.

Exam tip: background-color: yellow would colour the background, not fulfil a request for yellow text.

Q4 Concept map: issuing a library book

Question: Show the relationships involved in issuing a book, including availability and the maximum number a user can borrow.



Decision rule

Let n be the number of books currently borrowed and L the maximum borrowing limit.
Issue the requested book only if it is available AND n < L.

1. Identify the book and check availability.
2. Identify the member and check their current loans.
3. If unavailable, do not issue the book.
4. If the limit has been reached, do not issue another book.
5. Otherwise, record issue and due dates, mark the book as issued, and increase the member's loan count.

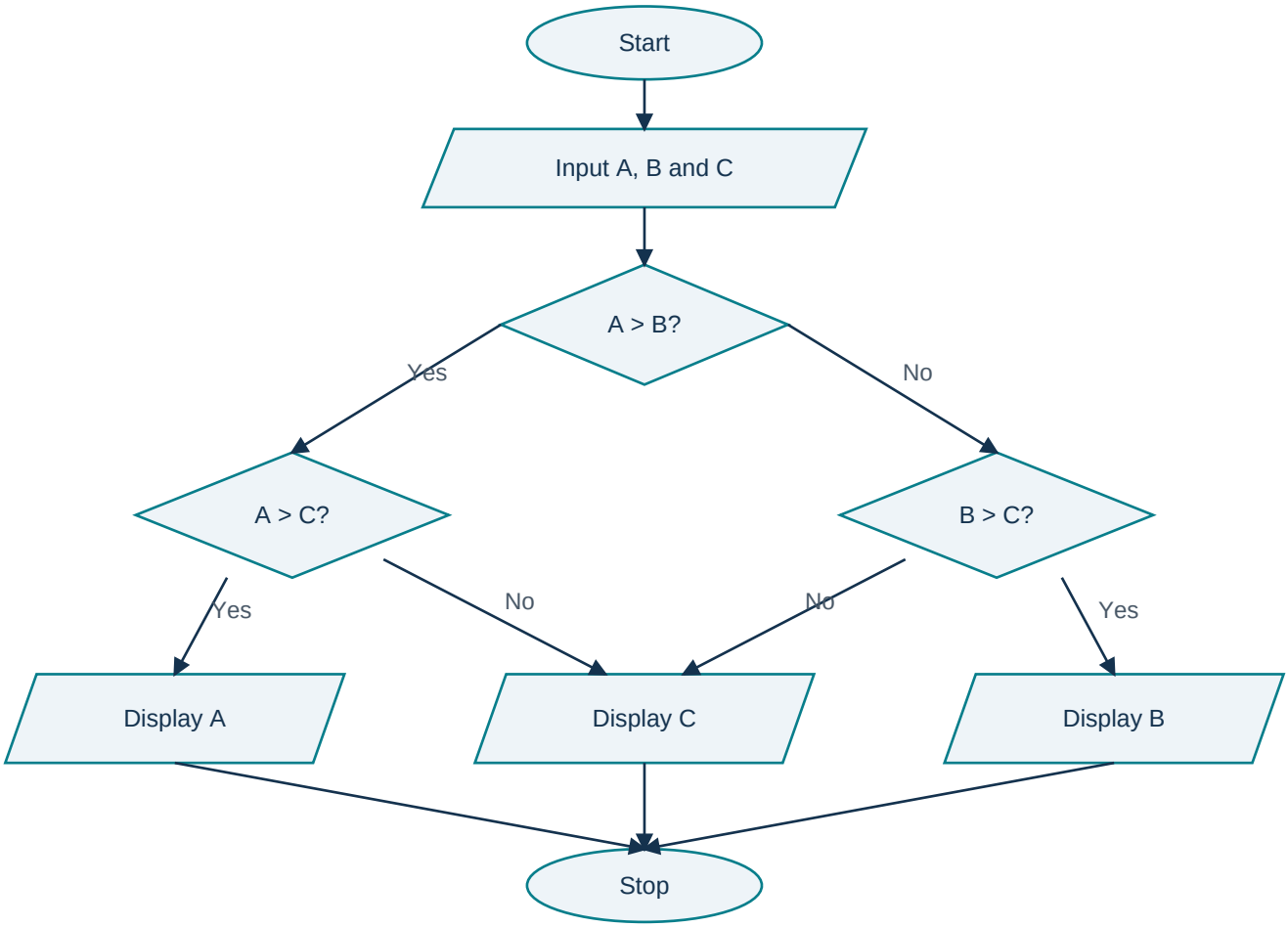
Exam tip: Use currentLoans < maximumLimit, not <=. For a limit of 3 books, a member with 2 can borrow one more; a member with 3 cannot. Do not invent a numerical limit when none is supplied.

EVENING GROUP

Q4 OR flowchart | Q5(a) device identification

Q4 - OR Largest of three unequal numbers

Question: Develop a flowchart to find the largest of three unequal numbers A, B and C.



Reason: If A > B, B cannot be largest, so compare A with C. Otherwise, because the numbers are unequal, B > A; compare B with C. Display the greater value from the final comparison.

Example: A = 12, B = 25, C = 18. A > B is No; B > C is Yes. Display **25**.

Exam tip: Use diamonds for comparisons, Yes/No labels on the branches, parallelograms for input/output and ovals for Start/Stop.

Q5(a) Label the devices

Label	Device	Identification / function
i	Mouse	Input device for pointer control and selection.
ii	Processor / CPU	Chip that executes instructions and processes data.
iii	Monitor	Output device displaying text, images and video.
iv	Flatbed scanner	Input device converting paper documents or pictures into digital form.
v	External hard drive	Portable external storage connected using a cable.

Exam tip: CPU refers to the processor chip shown in picture ii. The complete computer case is the system unit, not the CPU.

EVENING GROUP

Q5(b) classification | Q6 business plan | Final exam guidance

Q5(b) System software or application software?

Label / software	Classification	Reason
i. macOS	System software - operating system	Manages Mac hardware and applications.
ii. Microsoft Excel	Application software	Creates spreadsheets, calculations and charts.
iii. Google Chrome	Application software	Browses websites.
iv. AVG Antivirus	System software - utility	Detects and removes malware.
v. Linux	System software - operating system	Manages resources and supports applications.

Q6 Business plan: Skill Development Pakistan

Scenario: A web-based freelancing platform connects clients and freelancers, with payment support, transparent ratings, project management, service tracking and feedback. The response headings are not visible below the Evening scenario; the answer uses the Morning paper's business-plan structure.

Required component	Exam-ready answer
Company name	Skill Development Pakistan
Slogan	Connecting Skills with Opportunities.
Executive summary (one line)	Skill Development Pakistan connects clients with freelancers through a web platform offering project management, payment support, ratings and service tracking.
Objective 1	Create earning opportunities for skilled students and professionals by connecting them with clients.
Objective 2	Build trust and support successful projects through secure payment processes, transparent ratings and progress tracking.
Potential customer / market 1	Skilled students and professionals seeking freelance work.
Potential customer / market 2	Individuals and businesses seeking freelance services.

Reason: Freelancers provide services; clients purchase them. The plan addresses both user groups and matches the features given in the scenario. Payment methods mentioned in the scenario are scenario features, not a claim about their real-world availability.

Exam tip: Keep the exact company name, one concise summary, two separate objectives and both user groups. Use scenario details instead of writing an unrelated generic plan.

Final exam checklist

Use correct question numbers. State assumptions only where necessary. Write readable, indented code and close paired tags. For output questions, preserve text, values and line breaks. In diagrams, label branches and connect arrows. Practise every OR part, but attempt the required choice according to the actual exam instructions. These solutions support preparation; marks depend on the examination marking scheme.